

k-NN Graph Analysis of Stellar Spatial Data

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```
stars <- data.frame(  
  ra = runif(500, 0, 360),  
  dec = runif(500, -90, 90)  
)
```

Introduction

Understanding spatial structure in astronomical data is important for identifying patterns, clustering behavior, and connectivity among observed objects. In this project, I explore how local spatial structure emerges when stellar positions are represented as a k-nearest neighbor (k-NN) graph.

The guiding question for this analysis is:

How does local spatial structure emerge in stellar position data, and how does it depend on the choice of neighborhood size k in a k-nearest neighbor graph?

As a preliminary step, I use synthetic right ascension (RA) and declination (Dec) data to validate the graph construction and analysis pipeline before applying the method to real Gaia DR3 data.

Dataset Description

The dataset used in this initial stage is a synthetic dataset of 500 points representing stellar positions on the sky.

- **ra**: right ascension, measured in degrees from 0 to 360
- **dec**: declination, measured in degrees from -90 to 90

The synthetic dataset is used to establish a working analysis pipeline in a controlled setting.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is a critical first step in understanding the structure and characteristics of a dataset before applying more advanced analytical methods. In this project, EDA is used to examine the distribution and spatial behavior of synthetic stellar position data defined by right ascension (RA) and declination (Dec). This step helps establish expectations for how the data should behave and provides a baseline for interpreting graph-based results.

Summary Statistics

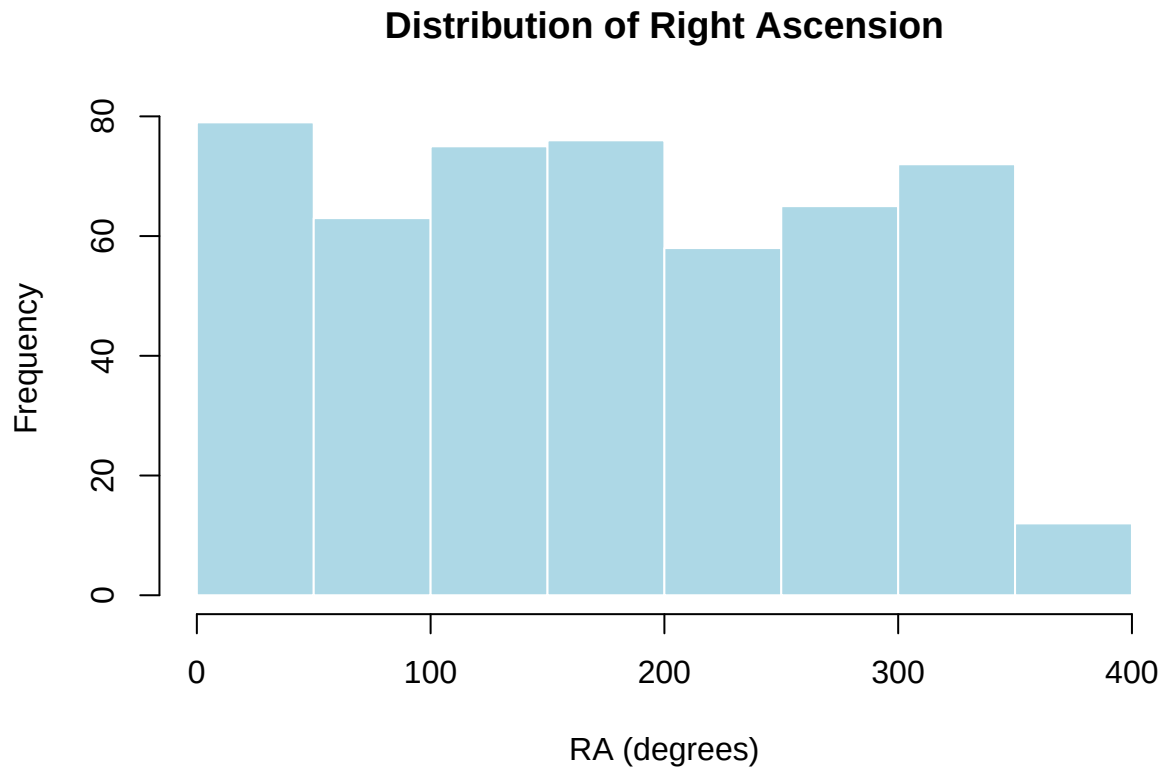
```
summary(stars)
```

The summary statistics provide a concise overview of the dataset. The right ascension (RA) values range from approximately 0 to 360 degrees, while declination (Dec) values range from approximately -90 to 90 degrees. These ranges are consistent with the definitions of celestial coordinates used in astronomy. The mean and median values for both variables fall near the center of their respective ranges, indicating that the data is evenly distributed. This is expected because the dataset was generated using uniform random sampling.

The absence of extreme skewness or outliers suggests that the dataset does not contain any inherent clustering or irregularities. This is important because it establishes that any structure observed later in the graph analysis is a result of the modeling approach rather than pre-existing patterns in the data.

Distribution of Right Ascension

```
hist(stars$ra,  
     main = "Distribution of Right Ascension",  
     xlab = "RA (degrees)",  
     col = "lightblue",  
     border = "white")
```

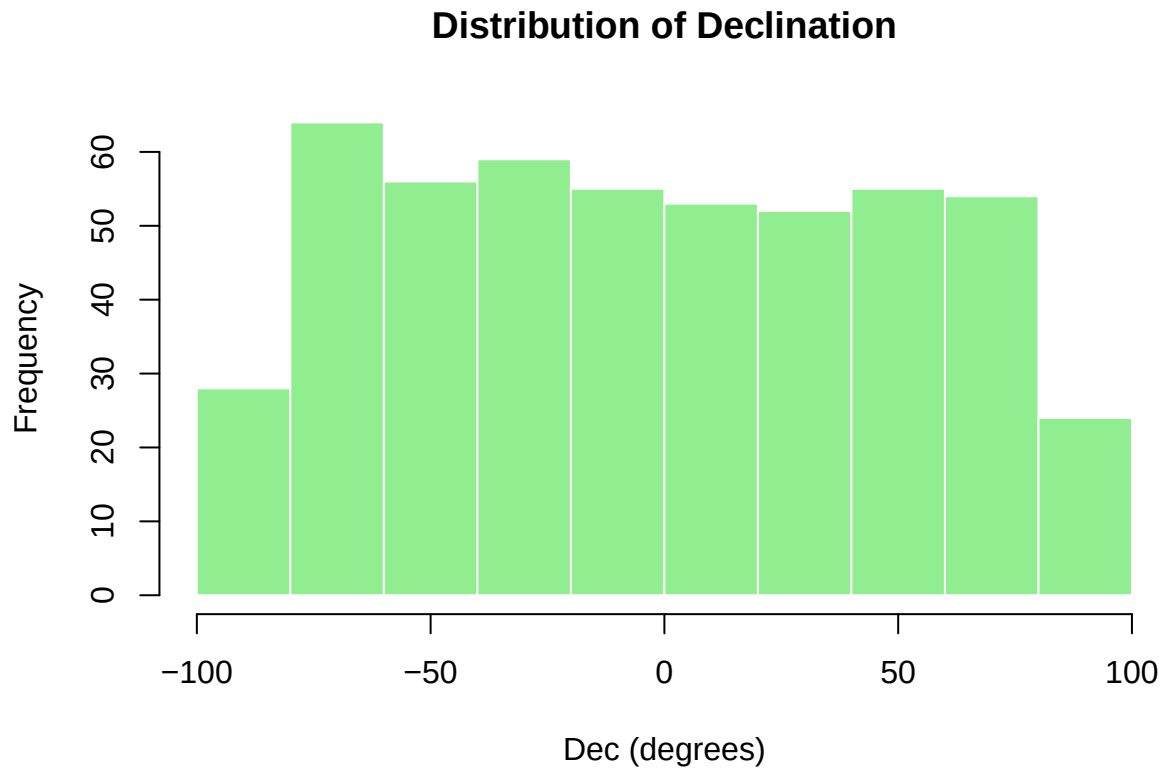


The histogram of right ascension values shows that the data is distributed relatively uniformly across the interval from 0 to 360 degrees. There are no visible peaks or concentrations, which indicates that no particular region of RA is favored in the dataset.

This uniform distribution is consistent with how the data was generated and confirms that the dataset does not contain directional bias. In a real astronomical dataset, one might expect to see clustering or density variations due to physical structures such as star clusters or the galactic plane. The absence of such features here reinforces that this dataset serves as a neutral baseline for testing the analysis pipeline.

Distribution of Declination

```
hist(stars$dec,  
     main = "Distribution of Declination",  
     xlab = "Dec (degrees)",  
     col = "lightgreen",  
     border = "white")
```



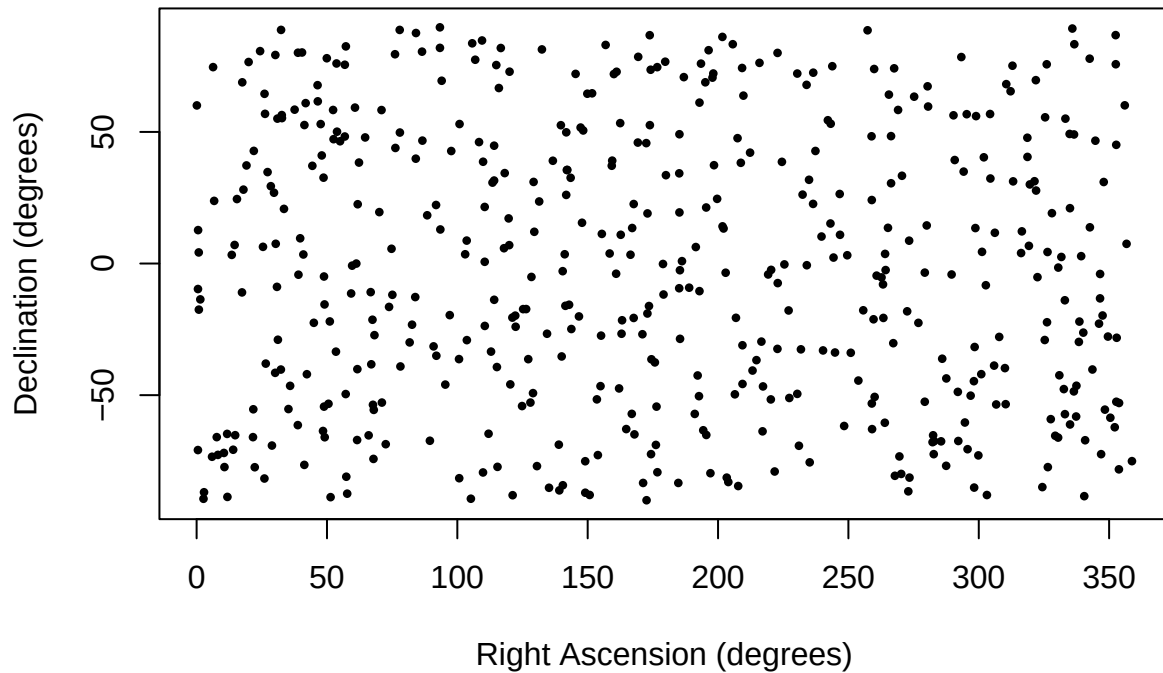
The histogram of declination values similarly shows a roughly uniform distribution across the range from -90 to 90 degrees. The values are spread evenly across both negative and positive regions, indicating symmetry about zero.

This symmetry is important because declination represents vertical positioning on the celestial sphere. The balanced distribution confirms that the dataset does not artificially favor the northern or southern celestial hemisphere. As with RA, the lack of clustering or irregularities confirms that the dataset is appropriate for baseline testing.

Spatial Distribution

```
plot(stars$ra, stars$dec,  
     pch = 16,  
     cex = 0.6,  
     main = "Spatial Distribution of Synthetic Stellar Positions",  
     xlab = "Right Ascension (degrees)",  
     ylab = "Declination (degrees)")
```

Spatial Distribution of Synthetic Stellar Positions

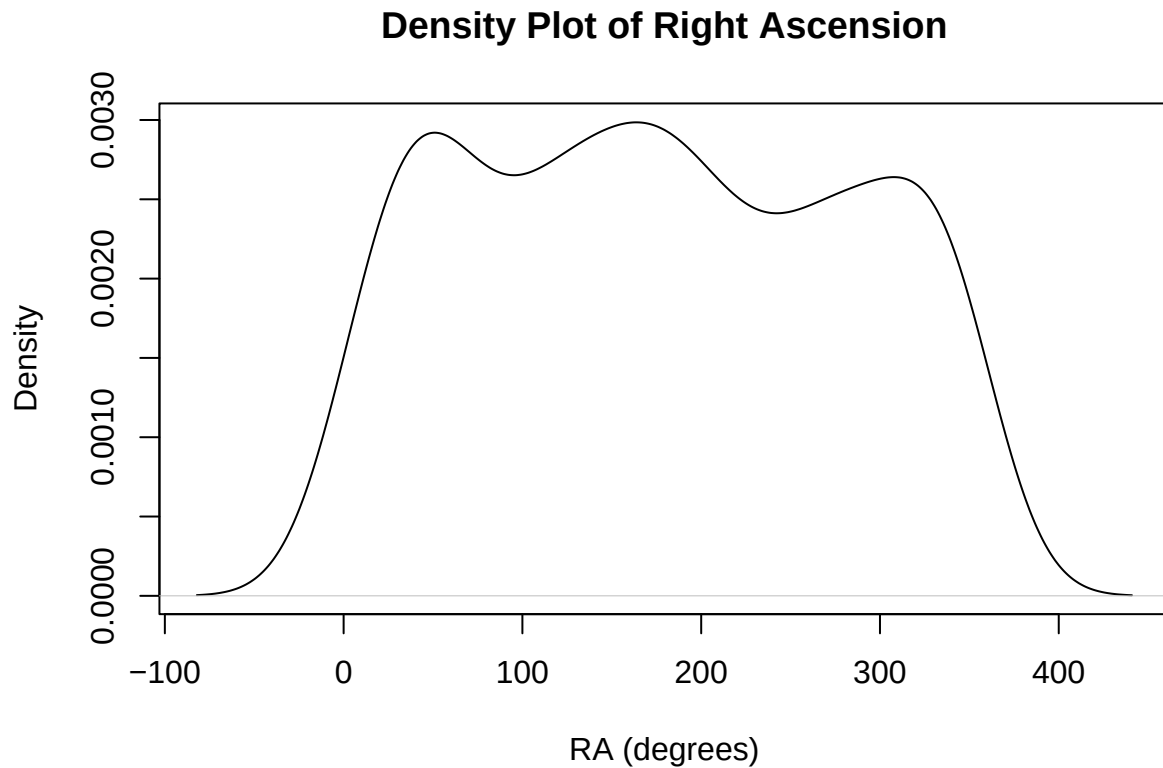


The scatter plot of right ascension versus declination provides a two-dimensional view of the spatial distribution of points. The points appear evenly scattered across the entire coordinate space, with no visible clusters or patterns.

This visualization is particularly important because it reflects the structure that will later be converted into a graph. The absence of dense regions or gaps suggests that the data represents a homogeneous spatial distribution. As a result, any connectivity or clustering observed in the k-NN graph will arise from the algorithm itself rather than underlying spatial structure.

Density of Right Ascension

```
plot(density(stars$ra),  
     main = "Density Plot of Right Ascension",  
     xlab = "RA (degrees)")
```

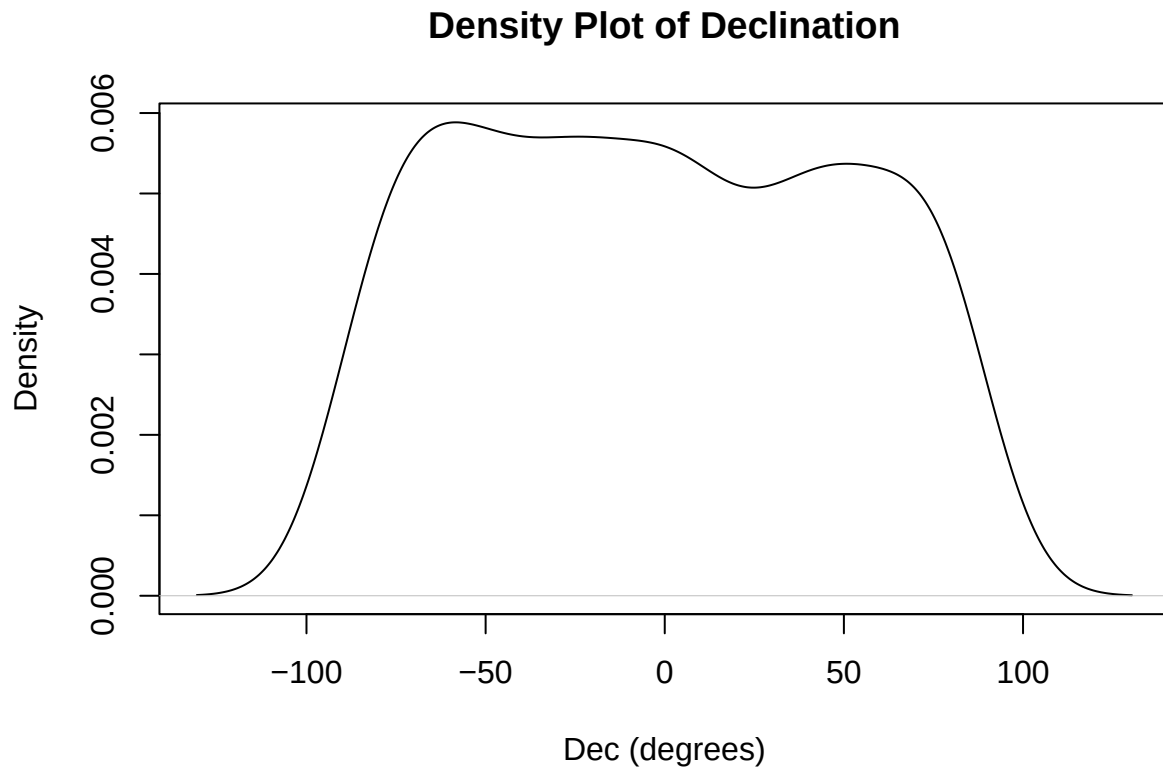


The density plot provides a smoothed estimate of the distribution of right ascension values. The curve is relatively flat, which is consistent with a uniform distribution. Unlike the histogram, which depends on binning, the density plot offers a continuous representation that confirms the absence of strong peaks or concentrations.

This further reinforces the conclusion that the dataset does not exhibit inherent clustering in the RA dimension.

Density of Declination

```
plot(density(stars$dec),  
     main = "Density Plot of Declination",  
     xlab = "Dec (degrees)")
```



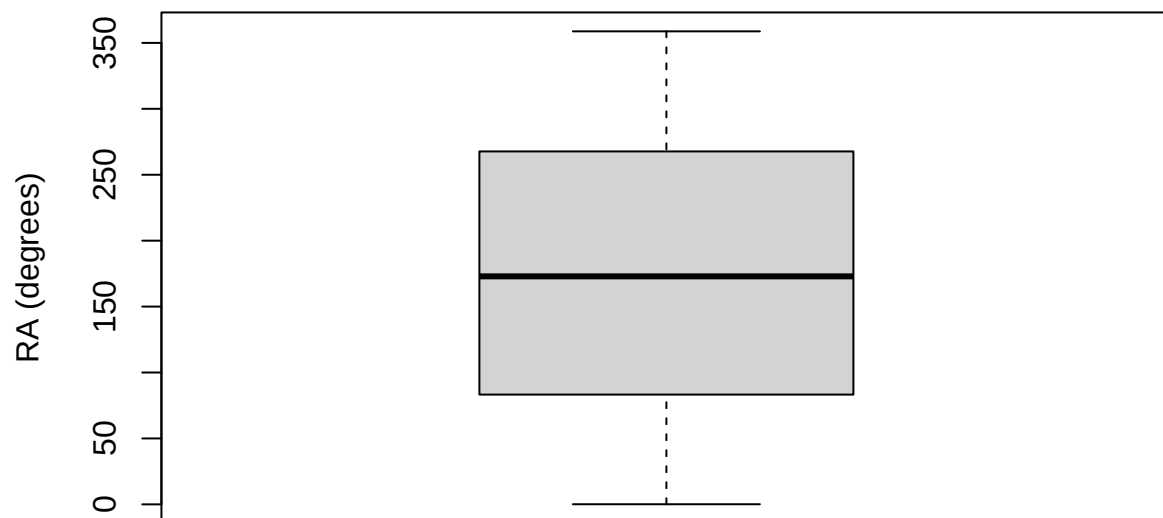
The density plot for declination also shows a smooth and relatively uniform distribution. There are no sharp peaks or irregularities, indicating that the values are evenly distributed across the full range.

Together, the density plots for RA and Dec confirm that the dataset behaves as expected under uniform random sampling. This consistency is important for validating the subsequent graph analysis.

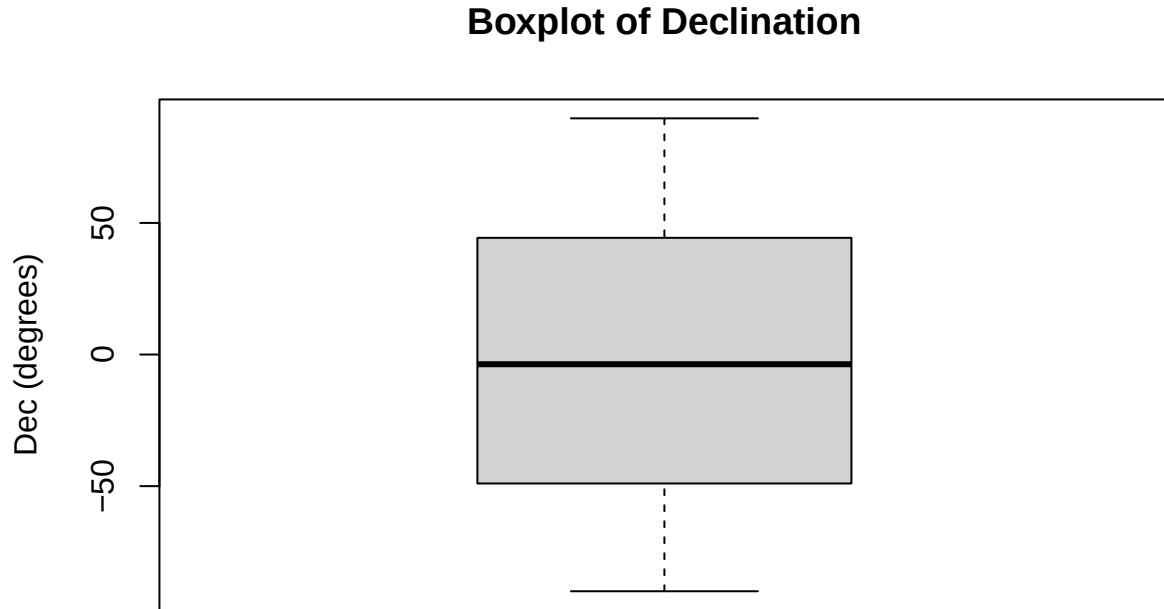
Boxplots of RA and Dec

```
boxplot(stars$ra,  
        main = "Boxplot of Right Ascension",  
        ylab = "RA (degrees)")
```

Boxplot of Right Ascension



```
boxplot(stars$dec,  
        main = "Boxplot of Declination",  
        ylab = "Dec (degrees)")
```



The boxplots provide a compact summary of the spread and central tendency of each variable. Both RA and Dec show a wide interquartile range and no significant outliers, which is consistent with a uniform distribution.

The median values lie near the center of the range, and the whiskers extend evenly in both directions. This indicates that the dataset is balanced and lacks extreme values or anomalies.

EDA Summary

Overall, the exploratory data analysis confirms that the synthetic dataset is uniformly distributed in both right ascension and declination. There is no evidence of clustering, skewness, or irregular spatial patterns. This makes the dataset well-suited as a baseline for testing graph-based methods.

Because the dataset lacks inherent structure, any patterns observed in the k-nearest neighbor graph can be interpreted as a result of the graph construction process rather than the underlying data. This provides a clear reference point for comparison when real Gaia data is introduced in later stages of the project.

k-NN Graph Analysis and Effect of k

The k-nearest neighbor (k-NN) graph is constructed by connecting each point in the dataset to its k closest neighbors based on Euclidean distance in the (RA, Dec) coordinate space. The choice of k plays a critical role in determining the structure of the resulting graph.

Small values of k tend to produce sparse graphs with potentially disconnected components, while larger values of k create denser graphs with higher connectivity. Understanding how graph properties change with k is essential for selecting an appropriate neighborhood size.

To investigate this, I analyze the graph structure across multiple values of k and compute key network metrics.

```
library(FNN)
library(igraph)

##
## Attaching package: 'igraph'

## The following object is masked from 'package:FNN':
##
##      knn

## The following objects are masked from 'package:stats':
##
##      decompose, spectrum

## The following object is masked from 'package:base':
##
##      union

coords <- as.matrix(stars)

k_values <- c(2, 5, 10, 15, 20)

results <- data.frame(
  k = numeric(),
  edges = numeric(),
  components = numeric(),
  avg_degree = numeric(),
  clustering = numeric()
)

for (k in k_values) {
```

```

knn <- get.knn(coords, k = k)

edges <- cbind(
  rep(1:nrow(coords), each = k),
  as.vector(knn$nn.index)
)

g <- graph_from_edgelist(edges, directed = FALSE)
g <- simplify(g)

results <- rbind(results, data.frame(
  k = k,
  edges = ecount(g),
  components = components(g)$no,
  avg_degree = mean(degree(g)),
  clustering = transitivity(g)
))
}

results

```

##	k	edges	components	avg_degree	clustering
## 1	2	995	1	3.980	0.005459509
## 2	5	2473	1	9.892	0.016639521
## 3	10	4890	1	19.560	0.035757471
## 4	15	7254	1	29.016	0.055749872
## 5	20	9575	1	38.300	0.075016362

Interpretation

The table above summarizes how key graph properties change as the number of neighbors k increases. Several clear trends emerge.

First, the number of edges increases with k , which is expected since each node is connected to more neighbors. As a result, the average degree of the graph also increases.

Second, the number of connected components decreases as k increases. For small values of k , the graph may consist of multiple disconnected clusters. However, as k grows, the graph quickly becomes fully connected. In this dataset, the graph becomes a single connected component at relatively low values of k , reflecting the uniform spatial distribution of points.

Finally, the clustering coefficient remains relatively low across all values of k . This is consistent with the lack of inherent structure in the synthetic dataset. Since points are randomly distributed, there are few tightly connected local neighborhoods.

Overall, these results show that increasing k strengthens connectivity but does not introduce meaningful clustering in uniformly distributed data. This highlights the importance of choosing k carefully, especially when working with real astronomical datasets where true spatial structure may exist. Unlike this synthetic dataset, real Gaia DR3 data is expected to exhibit non-uniform spatial distributions driven by astrophysical structures such as stellar clusters and the galactic plane.

k-Based Visual Comparisons

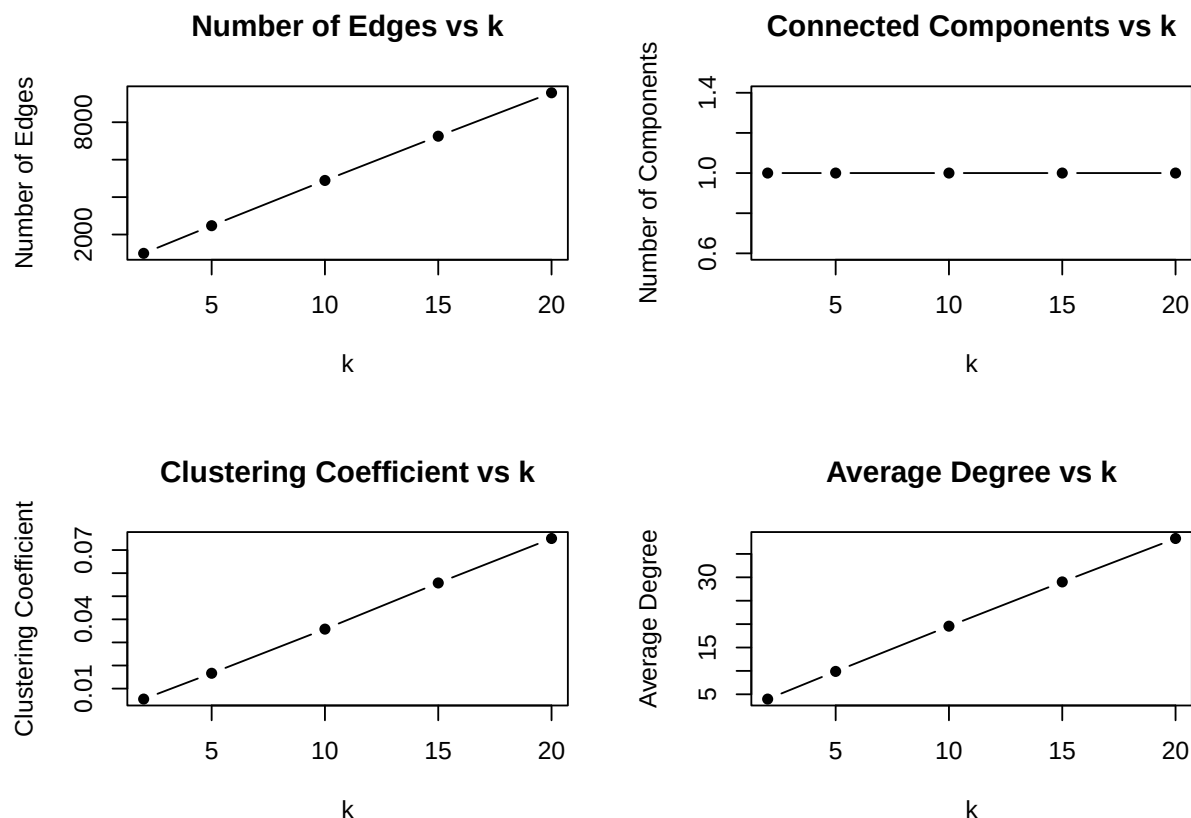
```
par(mfrow = c(2, 2))

plot(results$k, results$edges,
     type = "b", pch = 16,
     main = "Number of Edges vs k",
     xlab = "k", ylab = "Number of Edges")

plot(results$k, results$components,
     type = "b", pch = 16,
     main = "Connected Components vs k",
     xlab = "k", ylab = "Number of Components")

plot(results$k, results$clustering,
     type = "b", pch = 16,
     main = "Clustering Coefficient vs k",
     xlab = "k", ylab = "Clustering Coefficient")

plot(results$k, results$avg_degree,
     type = "b", pch = 16,
     main = "Average Degree vs k",
     xlab = "k", ylab = "Average Degree")
```



```
par(mfrow = c(1, 1))
```

The average degree increases approximately linearly with k , as expected, since each node connects to more neighbors. This confirms that the graph density is directly controlled by the parameter k .

Conclusion

In this project, I explored the construction and analysis of k -nearest neighbor graphs using synthetic stellar position data. The goal was to understand how spatial structure emerges and how it depends on the choice of neighborhood size k .

Using a uniformly distributed synthetic dataset, I established a baseline where no inherent clustering exists. The exploratory data analysis confirmed that the data is evenly distributed across both right ascension and declination, with no skewness or spatial bias.

The k -NN graph analysis showed that increasing k leads to greater connectivity, fewer connected components, and higher average degree. However, the clustering coefficient remained low, reflecting the lack of inherent structure in the data. This demonstrates that apparent structure in the graph can arise from the k -NN construction itself.

This work serves as a foundational step toward analyzing real Gaia DR3 data, where meaningful spatial patterns such as star clusters and galactic structures are expected. Future work will involve applying the same methods to real astronomical data and investigating how graph properties reflect physical phenomena.

Overall, this project demonstrates how apparent structure in the graph can arise from the k-NN construction itself, rather than underlying spatial patterns in the data.